# PRECOMPOSITION ON CONTINUOUS ALTERNATING MAPS IS NOT ANALYTIC IN CHARACTERISTIC 2

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Abstract. We study regularity of the map $\mathcal{L}(E,E') \to \mathcal{L}(\mathrm{Alt}^\iota(E',F),\mathrm{Alt}^\iota(E,F))$ given by precomposition, where $\mathrm{Alt}^\iota(E,F)$ denotes the space of continuous alternating $\iota$-linear maps between Banach spaces $E,F$. We construct a counterexample showing that this map is not always analytic when the field $k$ is complete non-Archimedean of characteristic 2 and not spherically complete. The result is formalized in Lean 4/Mathlib.

## 1. Introduction

Let $k$ be a nontrivially normed field. For normed spaces $E, E', F$ over $k$ and a finite $\iota$ we write $\mathcal{L}(E,E')$ for the space of continuous linear maps with the operator norm, and $\mathrm{Alt}^\iota(E,F)$ for the normed space of continuous alternating $\iota$-linear maps $E^\iota \to F$.

**Analytic functors.** A map $f\colon X \to Z$ between normed spaces over $k$ is **analytic at** $x_0 \in X$ if there are bounded $n$-linear maps $p_n\colon X^n \to Z$ and a radius $r > 0$ with $\sup_n \|p_n\| r^n < \infty$ such that $f(x_0+h) = \sum_{n\geq 0} p_n(h,\dots,h)$ whenever $\|h\| < r$. It is **analytic**, or $C^\omega$, if it is analytic at every point of $X$; this is the definition used in Mathlib [6].

Lang [5, Ch. III, §4] calls a functor $T$ on Banach spaces **of class $C^p$** if $f \mapsto T(f)$ is $C^p$. Bourbaki's *vector functors of class $C^r$* are defined in the same way [1, n° 7.6.1], and such a functor turns vector bundles into vector bundles [1, n° 7.6.2]. Similarly, we call a contravariant endofunctor $T$ on the category of normed spaces over $k$ **analytic** if for all $E, E'$ the map

$$\mathcal{L}(E,E') \to \mathcal{L}\big(T(E'),T(E)\big), \qquad f \mapsto T(f),$$

is analytic. For a normed space $F$ and a finite $\iota$, the assignment $E \mapsto \mathrm{Alt}^\iota(E,F)$, $f \mapsto (m \mapsto m\circ(f,\dots,f))$, is a contravariant functor. The assignment is also functorial in $F$: a continuous linear map $g\colon F \to F'$ acts by $m \mapsto g\circ m$. This action is linear in $g$, hence analytic, whereas the action of $f$ is not linear in $f$. So the bifunctor $\mathrm{Alt}^\iota$ is analytic if and only if the functor $\mathrm{Alt}^\iota(-,F)$ is analytic for every $F$.

Bourbaki introduces this bifunctor in [1, n° 7.8], as the functor $\alpha_d$ of continuous alternating $d$-linear maps with $\alpha_d(f_0,f_1)(u) = f_0 \circ u \circ (f_1,\dots,f_1)$, under the assumption that $k$ has characteristic 0 or that the bundles have finite rank, and states that it is not known whether this definition can be made without any restriction.

**Motivation: alternating bundles in Mathlib.** In Mathlib, manifolds and vector bundles are modelled on normed spaces (of possibly infinite dimension) over an arbitrary nontrivially normed field $k$, and their regularity is an element $n \in \mathbb{N}\cup\{\infty,\omega\}$. Differential forms require the bundle of continuous alternating maps between two $C^n$ vector bundles to be a $C^n$ vector bundle again (an instance of `ContMDiffVectorBundle`). Its transition functions are those

AI was used to assist in finding the result and in drafting this paper: the counterexample, its proof and the Lean formalization were developed with Claude, ChatGPT and the AutoformBot plugin.

of the given bundles with $\mathrm{Alt}^\iota$ applied, so this comes down to the regularity of the functor $\mathrm{Alt}^\iota(-,F)$, i.e. of the precomposition map

`ContinuousAlternatingMap.compContinuousLinearMapCLM`.

Two cases have been settled, in pull requests to Mathlib.

- If $|\iota|!$ is invertible in $k$ then $\mathrm{Alt}^\iota(-,F)$ is analytic, by polarisation. In Mathlib this is done in characteristic 0 by Gouëzel [2], who uses it to construct the bundle for every $n$ in characteristic 0 [3]; he remarks that a variant of the argument, not included there, covers finite-dimensional spaces in positive characteristic, but not the general case.
- Over every $k$, with no hypothesis on the characteristic or the dimension, $\mathrm{Alt}^\iota(-,F)$ is of class $C^n$ for every $n \in \mathbb{N} \cup \{\infty\}$ [7].

This leaves open the case where $|\iota|!$ is not invertible in $k$ and the spaces are infinite-dimensional: is $\mathrm{Alt}^\iota(-,F)$ analytic there? The following theorem provides a counterexample to the analyticity of the precomposition map.

**Theorem 1.** *Let $k$ be a complete non-Archimedean field of characteristic 2 which is not spherically complete. There is a Banach space $F$ over $k$ such that the functor $\mathrm{Alt}^2(-,F)$ is not analytic.*

## 2. Statement

**Notation.** For a polynomial $f$ and a monomial $\mu$, $\operatorname{coeff}_\mu f$ is the coefficient of $\mu$ in $f$. We write $c_0 \subset \ell^\infty$ for the null sequences, $\mathbf{1}_S$ for the indicator of a set $S$, and $\mathbf{1} = \mathbf{1}_{\mathbb{N}}$.

Throughout, $k$ is a field which is complete for a nontrivial absolute value $|\cdot|$ satisfying the **ultrametric inequality** $|x+y| \le \max(|x|,|y|)$; we call such a field **non-Archimedean**. The sup norms used below inherit the inequality. We assume that

(K1) $\operatorname{char} k = 2$;
(K2) $k$ is not spherically complete: there is a sequence of closed balls $B(c_n, r_n)$ such that $|c_{n+1} - c_n| \le r_n$, $r_0 > r_1 > \cdots$ and $\bigcap_n B(c_n, r_n) = \emptyset$.

The balls in (K2) are balls in $k$, and they are nested by the ultrametric inequality. Such fields exist; an explicit one is given in Appendix A. Let $E = \ell^\infty = \ell^\infty(\mathbb{N}, k)$ with the sup norm, a Banach algebra under pointwise multiplication; write $m_a \in \mathcal{L}(E,E)$ for multiplication by $a \in E$, so that $\|m_a\| = \|a\|$, and $e_i$ for the unit vectors. For $x, y \in E$ let

$$W(x,y) := x \otimes y + y \otimes x \in \ell^\infty(\mathbb{N}\times\mathbb{N}, k), \qquad W(x,y)_{pq} = x_p y_q + x_q y_p,$$

and let $F \subseteq \ell^\infty(\mathbb{N}\times\mathbb{N}, k)$ be the closed linear span of all $W(x,y)$, a Banach space. Let

$$A\colon \mathcal{L}(E,E) \to \mathcal{L}\big(\mathrm{Alt}^2(E,F), \mathrm{Alt}^2(E,F)\big), \qquad A(f)(m) = m \circ (f,f).$$

**Theorem 2.** *$A$ is not analytic at 0. In particular $A$ is not $C^\omega$.*

The map $A$ is the action on $\mathcal{L}(E,E)$ of the functor $\mathrm{Alt}^2(-,F)$, so Theorem 2 implies Theorem 1. In the terms of [1], $A$ is the map $f_1 \mapsto \alpha_2(\mathrm{id}_F, f_1)$. There class $C^r$ means analytic when $k \neq \mathbb{R}$, in a sense that agrees with ours for maps between Banach spaces, and norms over a non-Archimedean field are required to satisfy the ultrametric inequality, as those of $E$ and $F$ do. So Theorem 2 shows that over such a field $\alpha_2$ is not a vector functor of class $C^r$: the definition in [1, n° 7.8.1] cannot be made without any restriction.

Sections 3–6 prove Theorem 2 in four steps.

## 3. Step 1: analyticity forces a bilinear lift

Recall that $A\colon \mathcal{L}(E,E) \to \mathcal{L}(\mathrm{Alt}^2(E,F), \mathrm{Alt}^2(E,F))$ is given by $A(f)(m) = m \circ (f,f)$, so that $A(tf) = t^2 A(f)$. Steps 1 and 2 do not use (K2): until Step 3, $k$ may be any non-Archimedean field of characteristic 2.

**Lemma 3.** *Let $A\colon X \to Z$ be a map between normed spaces with $A(tf) = t^2 A(f)$. If $A$ is analytic at 0 then there is a bounded bilinear $P\colon X \times X \to Z$ with $P(f,f) = A(f)$ for all $f$.*

*Proof.* Let $\sum_n p_n$ be a power series of $A$ at 0, and fix $f \in X$. For scalars $t \in k$ with $|t|$ small,

$$t^2 A(f) = A(tf) = \sum_n t^n \, p_n(f, \dots, f).$$

Both sides are power series in the variable $t \in k$ with coefficients in $Z$. The coefficients of a convergent power series in one variable are unique over any nontrivially normed field, so comparing the coefficients of $t^2$ gives $p_2(f,f) = A(f)$. Take $P = p_2$. □

So assume that a bounded bilinear $P$ with $P(f,f) = A(f)$ exists. The map $W\colon E \times E \to F$ is bilinear, it is alternating because $W(x,x) = 2\,x \otimes x = 0$, and it is continuous because $\|W(x,y)\| \le \|x\|\|y\|$ by the ultrametric inequality; so $W \in \mathrm{Alt}^2(E,F)$ with $\|W\| \le 1$. Put, for $a, b, x, y \in E$,

$$\Phi(a,b;x,y) := P(m_a, m_b)(W)(x,y) \in F.$$

Then $\Phi$ is 4-linear, and since $\|m_a\| = \|a\|$ and $\|W\| \le 1$ we have

$$\|\Phi(a,b;x,y)\| \le C\|a\|\,\|b\|\,\|x\|\,\|y\|, \qquad C := \|P\|.$$

Here $C \ge 1$, because $P(\mathrm{id}, \mathrm{id}) = A(\mathrm{id})$ is the identity map of $\mathrm{Alt}^2(E,F) \neq 0$. Moreover

$$\Phi(a,b;x,x) = 0, \qquad \Phi(a,a;x,y) = W(ax, ay). \tag{1}$$

Since $2 = 0$, the first identity gives the symmetry $\Phi(a,b;x,y) = \Phi(a,b;y,x)$. We will contradict the existence of such a $\Phi$.

## 4. Step 2: a finite parity identity

Fix a finite set $I \subset \mathbb{N}$ of even size $2m$ and $a', b' \in E$ with $\|a'\|, \|b'\| \le 1$. Let $X_i, Y_i$ $(i \in I)$ be indeterminates. Expanding $\Phi$ multilinearly in $x = \sum_{i\in I} X_i e_i$ and $y = \sum_{i \in I} Y_i e_i$ defines, for any fixed $p, q \in I$, a polynomial

$$G_{pq} := \Phi(a' + x,\, b' + y;\, a' + x,\, b' + y)_{pq} \in k[X_i, Y_i]_{i \in I}$$

in the polynomial ring in these $4m$ variables, which we abbreviate to $k[X,Y]$. The values of the $G_{pq}$ at points $x, y \in k^I$ (vectors supported on $I$) are the entries of an $I \times I$ matrix $G(x,y)$. We also put

$$g_{pq} := \Phi(x,y;x,y)_{pq} = \sum_{i,j,k,l \in I} \Phi(e_i, e_j; e_k, e_l)_{pq}\, X_i Y_j X_k Y_l \in k[X,Y].$$

**Proposition 4.** *The matrix $G = (G_{pq})_{p,q \in I}$ is symmetric with zero diagonal: $G_{pq} = G_{qp}$ and $G_{pp} = 0$ in $k[X,Y]$.*

*Proof.* Every element of $F$ is a symmetric matrix with zero diagonal, since $W(x,y)_{pq} = W(x,y)_{qp}$ and $W(x,y)_{pp} = 2x_p y_p = 0$, and both properties pass to the closed span. The coefficients of $G_{pq}$ are the $(p,q)$ entries of values of $\Phi$, which lie in $F$. □

**Proposition 5.** *The polynomial $g_{pq}$ is homogeneous of degree 2 in the $X_i$ and of degree 2 in the $Y_i$, and $G_{pq} - g_{pq}$ has total degree $\le 3$. In particular $G_{pq}$ has total degree $\le 4$.*

*Proof.* By 4-linearity, $G_{pq}$ is a sum of 16 terms, according to whether each of the four slots is filled by the constant vector ($a'$ or $b'$) or by the variable one ($x$ or $y$), and the total degree of a term is the number of its variable slots. So $g_{pq}$, which has two slots with $x$ and two with $y$, is exactly the degree 4 part of $G_{pq}$, and $G_{pq} - g_{pq}$ has total degree $\leq 3$. □

**Proposition 6.** *For $p \neq q$ we have, in $k[X, Y]$,*

$$g_{pq}(Y, X) = g_{pq}(X, Y) + X_p^2 Y_q^2 + X_q^2 Y_p^2 \qquad \text{and} \qquad g_{pq}(X, X) = 0.$$

*Proof.* The field $k$ is infinite, since it has a nontrivial absolute value, and a polynomial over an infinite field which vanishes at every point is zero. So it suffices to check these identities at points $x, y \in k^I$. By the symmetry in the last two slots, $g(y, x) = \Phi(y, x; y, x) = \Phi(y, x; x, y)$, so by bilinearity in the first two slots and (1)

$$\begin{aligned} g(x, y) + g(y, x) &= \Phi(x + y, x + y; x, y) + \Phi(x, x; x, y) + \Phi(y, y; x, y) \\ &= W\big((x + y)x, (x + y)y\big) + W(x^2, xy) + W(xy, y^2) \\ &= \big[W(x^2, xy) + W(x^2, y^2) + W(xy, xy) + W(xy, y^2)\big] \\ &\quad + W(x^2, xy) + W(xy, y^2) \\ &= W(x^2, y^2), \end{aligned}$$

where the last step uses $2 = 0$ and $W(xy, xy) = 0$. This is the matrix whose $(p, q)$ entry is $x_p^2 y_q^2 + x_q^2 y_p^2$. Finally $g(x, x) = W(x^2, x^2) = 0$. □

Let $\tau$ be the automorphism of $k[X, Y]$ swapping $X_i \leftrightarrow Y_i$, and $\delta$ the substitution $Y_i := X_i$.

**Lemma 7** (balanced coefficients)**.** *Let $f \in k[X, Y]$ with $\tau f = f$. For every $S \subseteq I$ the coefficient of $\prod_{i \in S} X_i Y_i$ in $f$ equals the coefficient of $\prod_{i \in S} X_i^2$ in $\delta f$.*

*Proof.* The coefficient of $\prod_S X_i^2$ in $\delta f$ is the sum of the coefficients in $f$ of all monomials $X^\alpha Y^\beta$ with $\alpha + \beta = 2 \cdot \mathbf{1}_S$. The involution $(\alpha, \beta) \mapsto (\beta, \alpha)$ of this set of monomials preserves coefficients (as $\tau f = f$), and its only fixed point is $\alpha = \beta = \mathbf{1}_S$; all other terms cancel in pairs. Thus the coefficient of $\prod_S X_i^2$ in $\delta f$ is equal to the coefficient of $X^{\mathbf{1}_S} Y^{\mathbf{1}_S} = \prod_{i \in S} X_i Y_i$ in $f$. □

A **perfect matching** of $I$ is a partition of $I$ into $m$ unordered pairs $\{p, q\}$, its edges; for example $\{1, 2, 3, 4\}$ has the three perfect matchings 12|34, 13|24 and 14|23. Since $G_{pq} = G_{qp}$ (Proposition 4), the product of the $G_{pq}$ over the edges of a perfect matching is well defined.

**Proposition 8.** *Let $M$ be a perfect matching of $I$ and $\nu := \prod_{i \in I} X_i Y_i$. Then*

$$\operatorname{coeff}_\nu \prod_{\{p,q\} \in M} G_{pq} = 1.$$

*Proof.* Write $G_{pq} = g_{pq} + c_{pq}$, where $g_{pq}$ is homogeneous of total degree 4 and $c_{pq}$ has total degree $\leq 3$ (Proposition 5). Expanding the product over the $m$ edges of $M$ gives $\prod_M g_{pq}$ plus terms that contain at least one factor $c_{pq}$, and each such term has total degree $\leq 3 + 4(m-1) = 4m - 1$. The monomial $\nu$ has total degree $4m$, so only $\prod_M g_{pq}$ contributes to its coefficient:

$$\operatorname{coeff}_\nu \prod_M G_{pq} = \operatorname{coeff}_\nu \prod_M g_{pq}.$$

Put $g'_{pq} := g_{pq} + X_p^2 Y_q^2$. By Proposition 6,

$$\tau g'_{pq} = g'_{pq} \qquad \text{and} \qquad \delta g'_{pq} = X_p^2 X_q^2.$$

A product containing a factor $X_p^2 Y_q^2$ has no squarefree monomial, so replacing every $g_{pq}$ by $g'_{pq}$ does not change the coefficient of $\nu$. By Lemma 7, applied to the $\tau$-invariant product $\prod_M g'_{pq}$, this coefficient is the coefficient of $\prod_{i\in I} X_i^2$ in

$$\delta \prod_M g'_{pq} = \prod_{\{p,q\}\in M} X_p^2 X_q^2 = \prod_{i\in I} X_i^2,$$

which is exactly 1. □

**Lemma 9.** *Let $A$ be a symmetric matrix with zero diagonal over a commutative ring of characteristic 2, of even size $2m$. Then $\det A = \sum_M \prod_{\{p,q\}\in M} A_{pq}^2$, the sum being over perfect matchings; and the number of perfect matchings is $(2m-1)!!$.*

*Proof.* In characteristic 2 we have $\det A = \sum_\sigma \prod_i A_{i\sigma(i)}$. As $A$ is symmetric, $\sigma$ and $\sigma^{-1}$ give the same product, so the permutations with $\sigma \neq \sigma^{-1}$ cancel in pairs and only the involutions remain. An involution with a fixed point $i$ contributes the factor $A_{ii} = 0$. An involution without fixed points is a product of $m$ disjoint transpositions $(p\ q)$; the pairs $\{p,q\}$ form a perfect matching $M$, every perfect matching arises exactly once in this way, and therefore

$$\det A = \sum_M \prod_{\{p,q\}\in M} A_{pq}A_{qp} = \sum_M \prod_{\{p,q\}\in M} A_{pq}^2.$$

Finally, pairing the smallest index with one of the $2m-1$ others and continuing with the remaining indices shows that there are $(2m-1)(2m-3)\cdots 3\cdot 1 = (2m-1)!!$ perfect matchings. □

**Lemma 10** (hypercube sum)**.** *Let $R$ be a commutative ring of characteristic 2 and let $f \in R[Z_1,\dots,Z_N]$ have total degree $\le N$. Then $\sum_{z\in\{0,1\}^N} f(z)$ is the coefficient of $Z_1Z_2\cdots Z_N$ in $f$.*

*Proof.* By linearity we may take $f$ to be a monomial $Z^\alpha$. At a point $z \in \{0,1\}^N$ it takes the value 1 if $z_i = 1$ for all $i$ in the support of $\alpha$, and 0 otherwise; so the sum is $2^{N-|\operatorname{supp}\alpha|}$, which vanishes in $R$ unless every variable occurs in $Z^\alpha$. The only monomial of degree $\le N$ in which all $N$ variables occur is $Z_1\cdots Z_N$. □

**Corollary 11.** *We have*

$$\sum_{x,y\in\{0,1\}^I} \det G(x,y) = 1,$$

*and consequently there are $x, y \in \{0,1\}^I$ with $|\det G(x,y)| \ge 1$.*

*Proof.* Each $G_{pq}$ has total degree $\le 4$ (Proposition 5), so for a perfect matching $M$ the product $\prod_{\{p,q\}\in M} G_{pq}$ has total degree $\le 4m$. By Lemma 10 with $N = 4m$, the number of variables of $k[X,Y]$, and by Proposition 8,

$$\sum_{x,y\in\{0,1\}^I} \prod_{\{p,q\}\in M} G_{pq}(x,y) = \operatorname{coeff}_\nu \prod_{\{p,q\}\in M} G_{pq} = 1, \qquad \text{where } \nu = \prod_{i\in I} X_iY_i.$$

For each pair $x, y$ the matrix $G(x,y)$ is symmetric with zero diagonal (Proposition 4), so Lemma 9 applies to it. In characteristic 2 we have the Frobenius endomorphism $a \mapsto a^2$ of $k$.

Hence

$$\sum_{x,y\in\{0,1\}^I} \det G(x,y) = \sum_{x,y\in\{0,1\}^I} \sum_{M} \prod_{\{p,q\}\in M} G_{pq}(x,y)^2$$
$$= \sum_{M} \Big( \sum_{x,y\in\{0,1\}^I} \prod_{\{p,q\}\in M} G_{pq}(x,y) \Big)^2 = \sum_{M} 1^2 = (2m-1)!! = 1,$$

because $(2m-1)!!$ is odd. By the ultrametric inequality, $1 \le \max_{x,y\in\{0,1\}^I} |\det G(x,y)|$. □

## 5. Step 3: locality on $\ell^\infty$

This step uses (K2) and the completeness of $k$, which are standing hypotheses of §2, but not the characteristic. Recall (K2): there are centres $c_n \in k$ and radii $r_0 > r_1 > \cdots > 0$ with $|c_{n+1} - c_n| \le r_n$ such that no element of $k$ lies in all the balls $B(c_n, r_n) = \{a \in k : |a - c_n| \le r_n\}$. Recall also that $e_i \in \ell^\infty$ are the unit vectors. For $x \in \ell^\infty$ write $x|_{\ge N}$ for $x$ with its coordinates below $N$ set to 0.

**Proposition 12.** *A bounded linear functional $\varphi$ on $\ell^\infty$ with $\varphi(e_i) = 0$ for all $i$ is zero.*

*Proof.* Suppose $\varphi \neq 0$. Since $r_{n+1} < r_n$ we have $(r_{n+1}/r_n)\,\|\varphi\| < \|\varphi\|$, so by the definition of the operator norm there is $u_n \in \ell^\infty$ with $|\varphi(u_n)| > (r_{n+1}/r_n)\,\|\varphi\|\,\|u_n\|$. Put $s_n := (c_{n+2} - c_{n+1})\,u_n/\varphi(u_n)$. Then

$$\varphi(s_n) = c_{n+2} - c_{n+1} \qquad \text{and} \qquad \|s_n\| = |c_{n+2} - c_{n+1}|\,\frac{\|u_n\|}{|\varphi(u_n)|} \le r_{n+1} \cdot \frac{r_n}{r_{n+1}\|\varphi\|} = \frac{r_n}{\|\varphi\|}.$$

The difference $s_n - s_n|_{\ge n}$ is finitely supported, so by linearity $\varphi(s_n - s_n|_{\ge n}) = 0$; hence $\varphi(s_n|_{\ge n}) = c_{n+2} - c_{n+1}$. Moreover $\|s_n|_{\ge n}\| \le \|s_n\| \le r_n/\|\varphi\|$. Define $y := \sum_k s_k|_{\ge k}$ coordinatewise; the sum is finite in every coordinate $i$ because $s_k|_{\ge k}$ vanishes at $i$ when $k > i$. By the ultrametric inequality $\|y\| \le \max_k \|s_k\| \le r_0/\|\varphi\|$, so $y \in \ell^\infty$. Likewise $\|y - \sum_{k<n} s_k|_{\ge k}\| = \|\sum_{k\ge n} s_k|_{\ge k}\| \le \max_{k\ge n}\|s_k\| \le r_n/\|\varphi\|$. Hence, using the telescoping sum $\sum_{k<n} \varphi(s_k|_{\ge k}) = c_{n+1} - c_1$,

$$|\varphi(y) + c_1 - c_{n+1}| = \Big|\varphi\Big(y - \sum_{k<n} s_k|_{\ge k}\Big)\Big| \le \|\varphi\| \,\Big\|y - \sum_{k<n} s_k|_{\ge k}\Big\| \le \|\varphi\| \cdot \frac{r_n}{\|\varphi\|} = r_n.$$

Put $a := \varphi(y) + c_1 \in k$. We have just shown that $|a - c_{n+1}| \le r_n$, and $|c_{n+1} - c_n| \le r_n$ by (K2), so by the ultrametric inequality $|a - c_n| \le \max\big(|a - c_{n+1}|, |c_{n+1} - c_n|\big) \le r_n$ for every $n$. Thus $a \in \bigcap_n B(c_n, r_n)$, which contradicts (K2). □

This is one direction of van der Put's theorem [8, Théorème (3.2)]: $k$ is spherically complete if and only if $(\ell^\infty/c_0)' \neq 0$. The next lemma, the description of the dual of $\ell^\infty$, is [8, Théorème (4.1)]; see also [9, Ch. 4]. We include a proof.

**Lemma 13.** *Let $\varphi$ be a bounded linear functional on $\ell^\infty$. Then $\varphi(e_i) \to 0$ as $i \to \infty$, and $\varphi(x) = \sum_i x_i \varphi(e_i)$ for every $x \in \ell^\infty$. In particular $|\varphi(u)| \le \|u\| \sup_{i\ge N} |\varphi(e_i)|$ whenever $u = u|_{\ge N}$.*

*Proof.* Suppose that $\varphi(e_i) \not\to 0$. Then there are $\varepsilon_0 > 0$ and indices $i_0 < i_1 < \cdots$ with $|\varphi(e_{i_k})| \ge \varepsilon_0$. Let $T\colon \ell^\infty \to \ell^\infty$ be the operator with $T(x)(i_k) = x_k/\varphi(e_{i_k})$ and all other coordinates 0; it is linear and $\|T(x)\| \le \|x\|/\varepsilon_0$. So $\psi := \varphi \circ T$ is a bounded functional, and $\psi(e_k) = 1$ for all $k$ because $T(e_k) = e_{i_k}/\varphi(e_{i_k})$. Let sh be the right shift, $\mathrm{sh}(x)(0) = 0$ and $\mathrm{sh}(x)(i+1) = x(i)$. Then $\psi \circ (1 - \mathrm{sh})$ is a bounded functional with

$$\big(\psi \circ (1 - \mathrm{sh})\big)(e_k) = \psi(e_k) - \psi(e_{k+1}) = 1 - 1 = 0 \qquad \text{for every } k,$$

so it vanishes by Proposition 12. But $(1-\mathrm{sh})\mathbf{1} = e_0$, so $0 = \psi(e_0) = 1$, a contradiction.

Hence $x_i\varphi(e_i) \to 0$ for every $x \in \ell^\infty$. By the ultrametric inequality a difference of two partial sums of $\sigma(x) := \sum_i x_i\varphi(e_i)$ is bounded by its largest term, so the partial sums form a Cauchy sequence, and since $k$ is complete the series converges, with $|\sigma(x)| \le \|x\| \sup_i|\varphi(e_i)| \le \|\varphi\|\|x\|$. So $\sigma$ is a bounded functional with $\sigma(e_j) = \varphi(e_j)$ for all $j$, so $\varphi = \sigma$ by Proposition 12.

For the last assertion let $u = u|_{\ge N}$. Then $u_i = 0$ for $i < N$, so $\varphi(u) = \sum_{i\ge N} u_i\varphi(e_i)$. Every term of this series has absolute value at most $\|u\| \sup_{i\ge N}|\varphi(e_i)|$; by the ultrametric inequality so does every partial sum, and hence so does the limit $\varphi(u)$. □

**Proposition 14** (multilinear Phillips property)**.** *Let $\lambda\colon (\ell^\infty)^d \to k$ be a continuous multilinear map and $\varepsilon > 0$. There is $N$ such that $|\lambda(u_1,\dots,u_d)| \le \varepsilon$ whenever $\|u_l\| \le 1$ for all $l$ and $u_l = u_l|_{\ge N}$ for some $l$.*

We call such an $N$ a **threshold** for $\lambda$; every larger integer is then a threshold too.

*Proof.* Fix $\varepsilon$. By induction on $d$. For $d = 1$ choose $N$ with $|\lambda(e_i)| \le \varepsilon$ for $i \ge N$ and apply Lemma 13.

Let $d \ge 2$ and assume the statement for forms in fewer than $d$ variables. It suffices to show, for each $l$, that there is $N_l$ such that $|\lambda(u_1,\dots,u_d)| \le \varepsilon$ whenever $u_1,\dots,u_d$ lie in the unit ball of $\ell^\infty$ and $u_l = u_l|_{\ge N_l}$; then $\max_l N_l$ is a threshold. Without loss of generality, assume $l = 1$. Suppose, for the sake of contradiction, that for every $N$ there are $u_1,\dots,u_d$ in the unit ball with $u_1 = u_1|_{\ge N}$ and $|\lambda(u_1,\dots,u_d)| > \varepsilon$.

Step 1: *for every $N$ there are $i \ge N$ and $w_2,\dots,w_d$ in the unit ball with $|\lambda(e_i, w_2,\dots,w_d)| > \varepsilon$.*

For such a tuple $u_1,\dots,u_d$, Lemma 13 applied to the bounded functional $\lambda(\cdot, u_2,\dots,u_d)$ gives $|\lambda(u_1,\dots,u_d)| \le \sup_{i\ge N}|\lambda(e_i,u_2,\dots,u_d)|$, so one of the terms on the right is $> \varepsilon$.

Define

$$\nu_{l,i}(v_1,\dots,v_{d-1}) := \lambda(v_1,\dots,v_{l-1},e_i,v_l,\dots,v_{d-1})$$

and $\mu_i := \nu_{1,i}$. These are forms in $d-1$ variables, so the inductive hypothesis applies to them.

Step 2: *construct indices $j_1 < j_2 < \cdots$, integers $0 = M_0 < M_1 < \cdots$ and vectors $z_2^{(s)},\dots,z_d^{(s)}$ in the unit ball, supported in $[M_{s-1}, M_s)$ such that*

$$|\mu_{j_s}(z_2^{(s)},\dots,z_d^{(s)})| = |\lambda(e_{j_s}, z_2^{(s)},\dots,z_d^{(s)})| > \varepsilon. \tag{2}$$

Suppose that stages $1,\dots,s-1$ have been carried out. Let $K_s$ be a common threshold for the finitely many forms $\nu_{l,i}$ with $2 \le l \le d$ and $i < M_{s-1}$; for $s = 1$ we put $K_1 := 0$ and $j_0 := -1$. By Step 1 with $N = \max(K_s, j_{s-1}+1)$ there are $j_s \ge K_s$, with $j_s > j_{s-1}$, and $w_2,\dots,w_d$ in the unit ball with $|\mu_{j_s}(w_2,\dots,w_d)| > \varepsilon$. Let $M_s > M_{s-1}$ be a threshold for $\mu_{j_s}$, and put

$$z_l^{(s)} := w_l|_{\ge M_{s-1}} - w_l|_{\ge M_s}, \qquad \text{so that} \qquad w_l = h_l + z_l^{(s)} + t_l,$$

where $t_l := w_l|_{\ge M_s}$ and $h_l = \sum_{i<M_{s-1}} c_{l,i}e_i$ with $|c_{l,i}| \le 1$. Expand $\mu_{j_s}(w_2,\dots,w_d)$ multilinearly into $3^{d-1}$ terms. A term containing some $t_l$ has absolute value $\le \varepsilon$, because $M_s$ is a threshold for $\mu_{j_s}$. Consider a term containing no $t_l$ but some $h_l$, say in slot $l$. By linearity in slot $l$, the term equals

$$\sum_{i<M_{s-1}} c_{l,i}\,\lambda(e_{j_s},\dots,\underbrace{e_i}_{\text{slot } l},\dots) = \sum_{i<M_{s-1}} c_{l,i}\,\nu_{l,i}(e_{j_s},\dots).$$

The first entry is still $e_{j_s} = e_{j_s}|_{\ge K_s}$, because $j_s \ge K_s$, and $K_s$ is a threshold for every $\nu_{l,i}$ with $i < M_{s-1}$. So every summand has absolute value $\le \varepsilon$, and so does the term, by the

ultrametric inequality. As $|\mu_{j_s}(w_2,\dots,w_d)| > \varepsilon$ while all these terms have absolute value $\le \varepsilon$, the ultrametric inequality shows that the remaining term $\mu_{j_s}(z_2^{(s)},\dots,z_d^{(s)})$ satisfies (2).

Step 3: because the supports $[M_{s-1}, M_s)$ of the $z_l^{(s)}$ are pairwise disjoint, and each $z_l^{(s)}$ lies in the unit ball, the sum $u_l := \sum_s z_l^{(s)}$ is well defined with $\|u_l\| \le 1$. Let $\chi := \lambda(\cdot, u_2, \dots, u_d)$, a bounded functional. Fix $s$ and write $u_l = u_l^{<s} + z_l^{(s)} + u_l^{>s}$, where $u_l^{<s}$ is supported in $[0, M_{s-1})$ and $u_l^{>s}$ in $[M_s, \infty)$. Expand $\chi(e_{j_s}) = \mu_{j_s}(u_2,\dots,u_d)$ multilinearly. Exactly as before, with $K_s$ and $M_s$ as in stage $s$, the terms containing some $u_l^{>s}$ have absolute value $\le \varepsilon$, because $M_s$ is a threshold for $\mu_{j_s}$; and the terms containing no $u_l^{>s}$ but some $u_l^{<s}$ have absolute value $\le \varepsilon$, because $K_s$ is a threshold for every $\nu_{l,i}$ with $i < M_{s-1}$. The remaining term is $\mu_{j_s}(z_2^{(s)},\dots,z_d^{(s)})$. By (2) and the ultrametric inequality, $|\chi(e_{j_s})| > \varepsilon$ for every $s$. Since $j_s \to \infty$, this contradicts Lemma 13. □

This is where (K2) is used. The statement fails on $c_0$, where $x \mapsto \sum_i x_i$ is a bounded functional taking the value 1 at every $e_i$, and it fails on $\ell^\infty$ over a spherically complete field, where the Hahn–Banach theorem [4] gives a nonzero bounded functional vanishing on $c_0$.

## 6. Step 4: gluing

Recall from Step 1 that $\Phi$ is 4-linear with values in $F$ and that

$$\|\Phi(a,b;x,y)\| \le C\|a\|\,\|b\|\,\|x\|\,\|y\| \qquad \text{with } C \ge 1. \tag{3}$$

We now combine Corollary 11 (Step 2) with Proposition 14 (Step 3).

Choose inductively finite sets $I_1 < I_2 < \cdots$ in $\mathbb{N}$ (every element of $I_t$ is smaller than every element of $I_{t+1}$) with $|I_t| = 2t$ and vectors $a_t, b_t \in \{0,1\}^{I_t}$ as follows. We write $a^{\le t} \in \ell^\infty$ for the vector that agrees with $a_s$ on $I_s$ for every $s \le t$ and is zero elsewhere, and define $a^{<t}$, $b^{\le t}$ and $b^{<t}$ in the same way. Given the earlier stages, apply Corollary 11 with $a' = a^{<t}$ and $b' = b^{<t}$ to get $a_t, b_t$ with $|\det G_t| \ge 1$, where $G_t = \Phi(a^{\le t}, b^{\le t}; a^{\le t}, b^{\le t})|_{I_t \times I_t}$. Put $\varepsilon_t := \frac{1}{2}C^{-2t}$ and use Proposition 14 (with $d = 4$) for the finitely many entry forms $\Phi(\cdot)_{pq}$, $p, q \in I_t$, to get $N_t$ such that these forms are at most $\varepsilon_t$ on unit arguments one of which is supported in $[N_t, \infty)$. All later sets $I_s$, $s > t$, are chosen inside $[N_t, \infty)$.

Let $a, b \in \ell^\infty$ be defined by $a|_{I_t} = a_t$, $b|_{I_t} = b_t$ and zero elsewhere. Then $a = \mathbf{1}_A$ and $b = \mathbf{1}_B$ are indicator functions of two subsets of $\mathbb{N}$, so $m_a, m_b$ are coordinate projections. Let $\mathcal{O} := \Phi(a,b;a,b) \in F$. Fix $t$ and split $a = a^{\le t} + a^{>t}$ and $b = b^{\le t} + b^{>t}$; the vectors $a^{>t}$ and $b^{>t}$ are supported in the later sets $I_s$, $s > t$, hence in $[N_t, \infty)$. Expand $\mathcal{O}$ multilinearly into 16 terms. The term containing neither $a^{>t}$ nor $b^{>t}$ restricts to $G_t$ on $I_t \times I_t$; each of the other 15 terms has entries of absolute value $\le \varepsilon_t$ there, by the choice of $N_t$. So $\|\mathcal{O}|_{I_t\times I_t} - G_t\| \le \varepsilon_t$ by the ultrametric inequality. The entries of $\mathcal{O}$ and of $G_t$ have absolute value $\le C$ by (3), because $a$, $b$, $a^{\le t}$ and $b^{\le t}$ lie in the unit ball. Now expand $\det \mathcal{O}|_{I_t\times I_t} = \det(G_t + \Delta_t)$, where $\Delta_t := \mathcal{O}|_{I_t\times I_t} - G_t$, multilinearly in columns. One term is $\det G_t$. Every other term contains a column of $\Delta_t$, whose entries have absolute value $\le \varepsilon_t$, so its absolute value is at most $\varepsilon_t C^{2t-1} < 1 \le |\det G_t|$. So the ultrametric inequality gives

$$|\det \mathcal{O}|_{I_t\times I_t}| = |\det G_t| \ge 1 \qquad \text{for every } t.$$

On the other hand, every element of $F$ is a uniform limit of matrices of finite rank, because each $W(x,y)$ has rank $\le 2$. So there is a matrix $R$ of some finite rank $r$ with $\|\mathcal{O} - R\| \le \eta := (2C)^{-1}$. Put $D := \mathcal{O} - R$; its entries have absolute value $\le \eta$, and those of $R = \mathcal{O} - D$ have absolute value $\le C$. Let $t > r$ and expand $\det(R + D)$ on $I_t \times I_t$ multilinearly in columns, as before. A term with more than $r$ columns from $R$ vanishes, because these columns are linearly dependent.

Every other term has at least $2t-r$ columns from $D$, so its absolute value is at most $C^r\eta^{2t-r}$, since $\eta \le 1 \le C$. By the ultrametric inequality, $|\det \mathcal{O}|_{I_t\times I_t}| \le C^r\eta^{2t-r} = 2^{r-2t}C^{2(r-t)} < 1$, which contradicts the previous display. This proves Theorem 2. □

## Appendix A. An explicit field

The purpose of this appendix is to construct a field satisfying (K1) and (K2). Towards that end, let $k_0 = \mathbb{F}_2$ (or any field of characteristic 2). For $j \ge 0$ let $k_0((t^{1/2^j}))$ be the field of *formal* Laurent series $\sum_{n\ge N} a_n t^{n/2^j}$ ($N \in \mathbb{Z}$, $a_n \in k_0$) in the single variable $t^{1/2^j}$. These fields form an increasing chain, and we let

$$\mathcal{P} := \bigcup_{j\ge 0} k_0((t^{1/2^j})), \qquad |f| := 2^{-\operatorname{ord} f},$$

be the field of **2-power Puiseux series**. Here $\operatorname{ord} f$ is the minimum $q$ for which $\operatorname{coeff}_{t^q} f \neq 0$. We set $|0| = 0$ by definition. Let $k$ be the completion of $\mathcal{P}$ for this (non-Archimedean) absolute value. This is characteristic 2 since $k_0 \hookrightarrow \mathcal{P} \hookrightarrow k$.

**Proposition 15.** *$k$ is not spherically complete.*

*Proof.* We exhibit a sequence of balls as in (K2). Put

$$\gamma_m := 1 - 2^{-(m+1)}, \qquad c_n := \sum_{m=0}^{n} t^{\gamma_m} \in \mathcal{P}, \qquad r_n := 2^{-\gamma_{n+1}}.$$

*The balls have the form required in* (K2). Since $c_{n+1} - c_n = t^{\gamma_{n+1}}$ we have $|c_{n+1} - c_n| = 2^{-\gamma_{n+1}} = r_n$. Since the $\gamma_m$ are strictly increasing to 1, the $r_n$ are strictly decreasing, and $r_n > 2^{-1} = \frac{1}{2}$.

*The intersection is empty.* Suppose that $a \in k$ satisfies $|a - c_n| \le r_n$ for all $n$. As $\mathcal{P}$ is dense in $k$ there is $y \in \mathcal{P}$ with $|a - y| < \frac{1}{2}$, and then for every $n$

$$|y - c_n| \le \max\big(|y-a|, |a-c_n|\big) \le \max\big(\tfrac{1}{2}, r_n\big) = r_n. \tag{4}$$

Now $y \in k_0((t^{1/2^j}))$ for some $j$. The exponent $\gamma_j = (2^{j+1}-1)/2^{j+1}$ has an odd numerator, therefore $\gamma_j$ is not a multiple of $1/2^j$ and so $\operatorname{coeff}_{t^{\gamma_j}} y = 0$. For $n \ge j$ we have $2^{-\gamma_j} > 2^{-\gamma_{n+1}} = r_n$. Furthermore, the term $t^{\gamma_j}$ occurs in $c_n$, so

$$\operatorname{coeff}_{t^{\gamma_j}}(y - c_n) = 0 - 1 \neq 0, \qquad \text{hence} \qquad \operatorname{ord}(y-c_n) \le \gamma_j \quad \text{and} \quad |y - c_n| \ge 2^{-\gamma_j} > r_n,$$

which contradicts (4). □

## Appendix B. Formalization

The Lean 4/Mathlib development is at

https://github.com/Deicyde/alternating-analytic-counterexample

The library `AltCounter/` contains no `sorry`, and the only axioms it uses are the three standard ones (`propext`, `Quot.sound` and `Classical.choice`). Its headline theorem assumes that Mathlib's

`ContinuousAlternatingMap.compContinuousLinearMapCLM`

is $C^\omega$ for all nontrivially normed fields, normed spaces and finite index types, and derives `False`; a second theorem does the same assuming this only for complete non-Archimedean fields of characteristic 2, Banach spaces, $E' = E$ and $|\iota| = 2$. Both are restated in a Mathlib-only file `Challenge.lean` (with `sorry` in place of the proofs, as that format requires) and checked against `Solution.lean` by the `leanprover/comparator` tool.

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